MATH 112 - LECTURE 14 EXIT TICKET

Name: VK Nguyen

1. Suppose $a_n \to a$. Can the sequence have infinitely many terms outside the ball $B(a, \varepsilon)$ for some fixed $\varepsilon > 0$? Explain.

$\to B(a, \varepsilon)$

No. $|a - a_n| < \varepsilon$ means $a_n \in (a-\varepsilon, a+\varepsilon)$ if $n \geq n_0 \Rightarrow$ Terms that lie outside $B(a, \varepsilon)$ can be terms are $< n_0$, which are finite

2. True or false: If $(|a_n|)$ converges, then (a_n) converges. Explain or give a counterexample.

false. let $a_n = (-1)^n$ Then $|(-1)^n|$ converges to 1 while $(-1)^n$ alternates between 1 and -1

3. Suppose $a_n \to a$. If you remove the first 100 terms of the sequence, does the remaining sequence converge to a? Explain.

Yes. Cuz there are infinitely many terms so the first 100 terms are insignificant.

Ex. $a_n = \frac{1}{n}$ converges to 0

then $\frac{1}{101}, \frac{1}{102}, \frac{1}{103}, \ldots$ still converge to zero

Scanned with CamScanner

MATH 112 - LECTURE 13 EXIT TICKET

Name: Vic Nguyen

1. Let d(x,y) = |x-y| on ℝ. What is d(2,5)?

$d(2,5) = |2-5| = |-3| = 3$

2. In ℝ with the usual metric, what is the open ball B(-2,3)?

$B(z,r) = \{w \in S : d(w,z) < r\}$

$B(-2,3) = \{w \in \mathbb{R} : d(w,-2) < 3\}$

$|w-(-2)| < 3 \Leftrightarrow |w+2| < 3 \Rightarrow w+2 < 3, w+2 > -3$

$w < 1, w > -5 \Rightarrow (-5,1)$

3. Can a subset of ℝ be both open and closed with respect to some metric? Give an example.

Yes $\mathbb{R} \subseteq \mathbb{R}$ so it can be both open and closed w/ respect to same metric

Scanned with CamScanner

MATH 112 - LECTURE 15 EXIT TICKET

Name: MK Nguyen

1. Does the sequence $a_n = \dfrac{(-1)^n}{n}$ converge? Explain.

No. Since $(-1)^n$ alternates between -1 and 1

2. If neither (a_n) nor (b_n) converges, must $(a_n + b_n)$ also fail to converge? Explain or give a counterexample.

Yes. Since $a_n \to a$, $\forall n_1 \in \mathbb{N} : \forall n \geq n_1$, $|a_n - a| < \varepsilon/2$. Same for b_n

$\Rightarrow |a_n - a| + |b_n - b| < \varepsilon$

3. Given two sequences (a_n) and (b_n) such that $\lim_{n\to\infty} a_n = 0$, what can you say about $\lim_{n\to\infty} a_n b_n$?

$\lim_{n\to\infty} a_n b_n = 0$

Scanned with CamScanner

MATH 112 - LECTURE 12 EXIT TICKET

Name: Vic Nguyen

1. A student writes

$$1 = \sqrt{1} = \sqrt{(-1)^2} = \sqrt{(-1)(-1)} = \sqrt{-1}\sqrt{-1} = i \cdot i = i^2 = -1.$$

What went wrong in this argument?

$\sqrt{(-1)^2} = |-1|$,

2. Let $z \neq 0$. Why does dividing z by $|z|$ always give a complex number on the unit circle?

since $z/|z|$ has modulus 1 $\Rightarrow$ $z/|z|$ is a complex number on unit circle

$z/|z| = \cos\theta + i\sin\theta$

3. Suppose z and w are nonzero complex numbers with lengths $|z|$ and $|w|$ and angles θ and ψ. What are the length and angle of the product zw?

length : $|z| \cdot |w|$

angle : $\theta + \psi$

Scanned with CamScanner

MATH 112 - LECTURE 11 EXIT TICKET

Name: Vic Nguyen

1. What is the multiplicative inverse of i?

$i \otimes y = 1 \Rightarrow y = \frac{1}{i} = \frac{1 \cdot i}{i \cdot i} = \frac{i}{i^2} = \frac{i}{-1}$

$= -i \Rightarrow i \cdot (-i) = -i^2 \Rightarrow -i^2 = -(-1) = 1$

2. Let z = 2 + 3i. Write z as an ordered pair in $\mathbb{C} = \mathbb{R}^2$, and identify Re(z) and Im(z).

$\mathbb{C} = \mathbb{R}^2 = \{(2,3) : 2,3 \in \mathbb{R}\}$

$Re(2,3) = 2 \in \mathbb{R}$

$Im(2,3) = 3 \in \mathbb{R}$

3. Let $z = x + iy \in \mathbb{C}$. For which values of x and y is z^2 a real number? Explain.

$z = x + iy$ is $\bar{z} = x - iy$

$z \cdot \bar{z} = |z^2| = (x+iy)(x-iy) = x^2 + y^2$

Scanned with CamScanner

MATH 112 - LECTURE 10 EXIT TICKET

Name: Vic Nguyen

1. **Suppose u is an upper bound of S. Why does showing that every number smaller than u fails to be an upper bound prove that u = sup S?**

Because it shows that only u can be the upper bound

2. **If u = sup S, does u have to belong to S? If not, how can u still be the supremum?**

No. Even if it's not in the set it can still be very close w/ an element in the set

3. **Why is the order of the quantifiers "for every ε > 0, there exists x ∈ S" important in the characterization of the supremum?**

Change order will change meaning. No matter how small $\varepsilon < 0$, $x \in S$ in the interval immediately to the left of u

4. **Let x, y ∈ ℝ with x < y. Are there infinitely many rational numbers r ∈ ℚ such that x < r < y? Explain why.**

Yes.

Ex: $x = 1$, $y = 2$

$1 < 1.1 < 2$

$1 < 1.11 < 2$

$1 < 1.111 < 2$

$1 < 1.111\ldots < 2$

Scanned with CamScanner

MATH 112 - LECTURE 9 EXIT TICKET

Name: VK Nguyen

1. Let $S = \{ -1/n : n \in \mathbb{N} \}$.

Find sup S. Does S have a maximum? Explain briefly.

sup $S = 0$, but $0 \notin S$ $\Rightarrow$ No

2. Let A be a nonempty bounded subset of $\mathbb{R}$ such that $\inf A = \sup A$.

What can you conclude about A? Explain briefly.

then $u^* = l^*$, so A has one element

3. State the completeness axiom for $\mathbb{R}$.

there exists a nonempty subset S of R that is upper bounded but has no supremum.

Scanned with CamScanner